

Aaysha Ali

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ACADEMIC OBJECTIVE

Undergraduate in Pharmaceutical Sciences (B.Pharm) at Jamia Hamdard, maintaining continuous academic distinction. Experienced in computer-aided drug design, molecular docking workflows, and analytical pharmacology. Seeking admission to a research-intensive Master's program and a graduate research assistantship to investigate target-ligand interactions, computational pharmacology, and novel therapeutic interventions.

EDUCATION

Jamia Hamdard New Delhi, Delhi, India
Bachelor of Pharmacy (B.Pharm) | Maintained Continuous Academic Distinction 2023 to Present

RESEARCH AND PUBLICATIONS

- B.Pharm Thesis Research** | Computational Pharmacology 2025 to Present
- Investigating *in silico* comparative molecular docking and ADMET profiling of novel ketamine analogues targeting human NMDA receptors for treatment-resistant depression.
 - Retrieved crystallographic data from the Protein Data Bank (PDB), prepared receptor-ligand complexes, and analyzed channel pore interactions to benchmark binding affinity.
- Peer-Reviewed Literature Review** | Zenodo Repository October 2025
- Authored a review paper evaluating the pharmacological properties and nutraceutical profile of *Tinospora cordifolia* (DOI: 10.5281/zenodo.19390923).
 - Extracted and consolidated quantitative bioactive data to evaluate safety thresholds and therapeutic indices from peer-reviewed pharmacology literature.
- Structure-Based CADD Pipeline (3HS4)** | The Insilico Lab Fellowship July 2026
- Executed a five-ligand structure-based drug design workflow targeting Human Carbonic Anhydrase II (PDB: 3HS4): drug-likeness screening via SwissADME and protein preparation via UCSF Chimera.
 - Mapped binding pockets via DoGSiteScorer, profiled pharmacokinetics with ADMET-AI, and evaluated structural alignments and RMSD against AlphaFold3 predictions using PyMOL.
- Conference Poster Presentation** | SECON 2025, Jamia Hamdard October 2025
- Presented an ethnopharmacological research poster at the 12th SFE-India International Convention on Traditional Medicine.
 - Defended mechanistic findings and clinical implications of natural bioactive compounds before academic evaluators and international delegates.

TECHNICAL TRAINING AND CERTIFICATIONS

Verification links and credential archive: github.com/aaysha16/Certificates

- Good Clinical Practice (GCP)** June 2026
NIDA Clinical Trials Network (6-Hour course; Version 6, valid through June 2029)
- Molecular Docking: From Theory to Application** June 2026
Discovery Boulevard (Virtual screening, AutoDock/Vina, grid generation, PyMOL interaction modeling)
- HPLC Industrial Training and Laboratory Visit** March 2026
Arbro Pharmaceuticals Pvt. Ltd. Analytical Division (In collaboration with IPA-Delhi State Branch)

SCHOLARSHIPS AND AWARDS

- Pharmaceutical Industry Merit Scholarship** New Delhi, Delhi, India
Smile Foundation 2023 to Present
- Secured competitive merit-based scholarship; maintained across all semesters through consistent academic distinction.

TECHNICAL AND RESEARCH SKILLS

- Computational Drug Discovery:** AutoDock, AutoDock Vina, PyMOL, BIOVIA Discovery Studio, UCSF Chimera, SwissADME, ADMET-AI, AlphaFold3, ChemDraw.
- Analytical Instrumentation:** High-Performance Liquid Chromatography (HPLC) sample preparation and column equilibration, UV-Vis Spectroscopy, basic extraction protocols.
- Research Methodology:** Good Clinical Practice (GCP) guidelines, clinical trial protocol evaluation, scientific literature synthesis, PubMed, PubChem, PDB, Mendeley, Python, GraphPad Prism.
- Languages:** English (fluent, academic writing), Hindi (proficient), Urdu (native).